

# MEASURING CHAMPIONSHIP VENUE DIFFICULTY IN U.S. OPEN GOLF

*A Cross-Dataset Analysis Using Adjusted Field Scoring Average, Formal Inferential Tests, Women's Open Results, and Amateur Stroke Play Data*

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**PREPRINT STATUS:** This is a preprint. The manuscript has been submitted to the Journal of Quantitative Analysis in Sports (JQAS) and has not yet completed peer review. This version has not been peer reviewed.

**SUGGESTED CITATION:** Mohiuddin, J. (2026). *Measuring championship venue difficulty in U.S. Open golf: A cross-dataset analysis using adjusted field scoring average, formal inferential tests, women's open results, and amateur stroke play data [Preprint]*. SportRxiv.

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Submitted for peer review — July 2026 · ORCID: <https://orcid.org/0009-0009-8806-4853>

*Keywords:* U.S. Open Golf, venue difficulty, golf analytics, course architecture, sport statistics, championship setup

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## ABSTRACT

Conventional assessments of U.S. Open venue difficulty — winning score or raw field scoring average versus par — are confounded by USGA setup decisions, weather variation, field strength, and equipment technology. This paper proposes adjusted field scoring average (FSA+/-), anchored to era-specific field quality, as the primary difficulty measure, and introduces formal inferential testing to validate a four-tier venue classification. Welch's two-sample t-test confirms high-difficulty venues

(Tier 1+2) produce significantly higher FSA+/- than lower-difficulty venues:  $t(14.5) = 3.408$ ,  $p = 0.0041$ , Cohen's  $d = 1.558$ . One-way ANOVA confirms the four-tier framework is highly significant:  $F(3,13) = 12.001$ ,  $p = 0.0005$ ,  $\eta^2 = 0.735$ . The 2021 U.S. Amateur at Oakmont — 5.47 strokes harder than its co-host on the same day with no Open setup ( $t \approx 9.851$ ,  $p < 0.0001$ ,  $d \approx 1.577$ ) — provides causal evidence that Oakmont's difficulty is architectural. In the modern era, Merion 2013 (+4.55) is the hardest Open by adjusted field average. A statistically validated four-tier venue classification is presented.

## 1. INTRODUCTION

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The U.S. Open Championship is widely considered the most demanding test of golf among the four major championships. The USGA explicitly designs its course setup to identify the best player in the field under conditions of maximum difficulty. However, the difficulty of a given Open is a product of at least two distinct factors: the intrinsic architectural properties of the host venue, and the exogenous setup decisions imposed by the USGA for that specific week.

Golf analytics discourse has historically conflated these two sources of difficulty. Post-tournament assessments typically rely on winning score or raw field scoring average versus par — metrics that are demonstrably inadequate for cross-year venue comparison. The winning score reflects one outlier player's performance; Tiger Woods winning Pebble Beach 2000 by fifteen shots tells us almost nothing about that course's inherent difficulty. Field scoring average is better, but remains confounded by weather, era-specific field quality, and USGA setup decisions that vary across visits to the same venue. Neither metric has been subjected to formal inferential testing in the published literature.

This paper makes four contributions. First, it operationalizes adjusted field scoring average as an inflation-adjusted cross-era difficulty measure. Second, it introduces formal statistical testing — Welch's t-tests, one-way ANOVA, and a quasi-experimental natural experiment design — to validate venue difficulty classifications. Third, it applies a cross-dataset validation methodology using four independent data sources at shared venues. Fourth, it produces a statistically validated four-tier venue classification for all U.S. Open sites with sufficient data.

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## 2. PRIOR LITERATURE AND METHODOLOGICAL CONTEXT

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### 2.1 Strokes Gained and Adjusted Scoring

The strokes gained methodology, developed by Broadie (2012) and operationalized through the PGA Tour's ShotLink system, provides the most rigorous existing framework for performance measurement in golf. At the tournament level, strokes gained theoretically provides a clean difficulty signal. However, granular ShotLink data is not publicly available in a form permitting cross-tournament difficulty comparisons across U.S. Open venues and years. Data Golf's verified field scoring averages for 2010–2026 serve as the closest publicly available approximation.

### 2.2 Course Rating and Architectural Analysis

The USGA Course Rating system provides a standardized measure of expected score for a scratch golfer under normal conditions. However, Course Rating is calibrated to normal playing conditions rather than U.S. Open setup, and does not capture the variance introduced by championship-specific modifications. Qualitative architectural analyses (Doak, 1992; Cornish and Whitten, 1993; Hurdzan, 2004) provide frameworks for assessing difficulty through design features but lack the quantitative grounding necessary for cross-venue comparison.

### 2.3 The Era-Adjustment Problem and the Inflation Analogy

A fundamental challenge in cross-era comparison is the non-stationarity of field quality. A field scoring average of +7.23 in 1951 against a mixed field is not directly comparable to +3.82 in 2025 against the world's 156 best players. We address this with a framework analogous to inflation adjustment in macroeconomics: for within-era comparisons (2010–2026), Data Golf's figures are treated as approximately era-adjusted; for cross-era comparisons, pre-modern figures are reported as directional indicators only — nominal rather than real — and excluded from formal inferential tests.

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## 3. DATA, STATISTICAL METHODOLOGY, AND ANALYTICAL FRAMEWORK

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### 3.1 Primary Dataset: Modern Era FSA+/-

The primary difficulty measure is field scoring average versus par (FSA+/-), defined as the mean score across all field members per round, averaged across all four rounds, expressed relative to

course par. For 2010–2026, FSA+/- figures are drawn from Data Golf's publicly available championship database. For historical events (1947–2009), figures are sourced from GOLF Magazine retrospective rankings, ESPN Stats and Information, and USGA championship records. A critical caveat applies to 2018 Shinnecock Hills (+4.39): the Saturday afternoon round was halted and greens were watered mid-competition, artificially inflating the official figure. We estimate the true architectural FSA+/- at approximately +3.6 to +3.8.

Descriptive statistics for the  $n = 17$  modern era observations (2010–2026): mean  $\mu = 2.955$ , SD = 1.075, SE = 0.261, median = 3.090, minimum = 1.140 (Erin Hills, 2017), maximum = 4.550 (Merion East, 2013). The distribution is approximately symmetric with moderate left skew attributable to three anomalously easy Opens (Erin Hills 2017, Pebble Beach 2019, LACC 2023), all producing winning scores of  $-10$  or lower.

**Table 1. Descriptive Statistics — Modern Era FSA+/- (2010–2026,  $n = 17$ )**

| Statistic               | Value | Statistic | Value |
|-------------------------|-------|-----------|-------|
| Mean ( $\mu$ )          | 2.955 | Median    | 3.090 |
| Standard deviation (SD) | 1.075 | Minimum   | 1.140 |
| Standard error (SE)     | 0.261 | Maximum   | 4.550 |
| Sample size ( $n$ )     | 17    | Range     | 3.410 |

*FSA+/- = field scoring average vs. par. Source: Data Golf (datagolf.com), 2010–2026.*

### 3.2 Cross-Validation Datasets

**Dataset 2 — U.S. Women's Open:** The Women's Open has been contested at several venues that have also hosted the Men's Open. We use the 36-hole cut score versus par as the Women's Open difficulty proxy, sourced from USGA Championship Records and Wikipedia.

**Dataset 3 — U.S. Amateur (Men's) Stroke Play:** The Amateur's stroke play qualifying round provides the cleanest architectural difficulty signal in the dataset, as it is contested without Open-specification rough ( $\approx 1.5$  inches vs.  $\approx 3.5$ – $4.0$  inches for the Open) and without punishing pin positions. Data sourced from USGA Media Center press releases, Golf Digest, and the Golf Bible historical database.

**Dataset 4 — U.S. Women's Amateur:** Additional cross-gender architectural evidence at shared venues, sourced from USGA records.

### 3.3 Formal Statistical Tests

#### 3.3.1 Welch's Two-Sample t-Test: High- vs. Lower-Difficulty Venue Groups

To test whether proposed Tier 1+2 venues produce significantly higher FSA+/- than Tier 3+4 venues, we apply Welch's two-sample t-test, which does not assume equal population variances — an appropriate choice given the heterogeneity of the two groups (Ruxton, 2006). Null hypothesis  $H_0$ :  $\mu_{\text{(Tier 1+2)}} = \mu_{\text{(Tier 3+4)}}$ ; alternative  $H_1$ :  $\mu_{\text{(Tier 1+2)}} > \mu_{\text{(Tier 3+4)}}$ . Tier 1+2 comprises Tier 1 (Oakmont CC  $\times 2$ , Winged Foot West) and Tier 2 (Merion East, Olympic Club Lake),  $n = 5$  modern observations. Tier 3+4 comprises all remaining venues,  $n = 11$ . Significance level  $\alpha = 0.05$ .

#### 3.3.2 One-Way ANOVA: FSA+/- Across All Four Tiers

To assess whether mean FSA+/- differs systematically across all four proposed venue tiers, we apply a one-way analysis of variance (ANOVA). The omnibus F-test evaluates  $H_0$ :  $\mu_1 = \mu_2 = \mu_3 = \mu_4$  against  $H_1$ : at least one tier mean differs. Group sizes:  $n_1 = 3$  (Tier 1),  $n_2 = 2$  (Tier 2),  $n_3 = 7$  (Tier 3),  $n_4 = 4$  (Tier 4). Effect size is reported as eta-squared ( $\eta^2$ ).

#### 3.3.3 Natural Experiment: 2021 U.S. Amateur at Oakmont (Causal Identification)

The 2021 U.S. Amateur co-host design permits a quasi-experimental causal estimate of Oakmont Country Club's venue-specific difficulty premium. The 312-player field was randomly assigned to Round 1 at either Oakmont Country Club ( $n \approx 78$ ) or co-host Longue Vue Club ( $n \approx 78$ ), with venues reversed for Round 2. Random assignment eliminates selection bias. We apply Welch's two-sample t-test to the published round averages (Oakmont:  $\bar{x} = 77.16$ ; Longue Vue:  $\bar{x} = 71.69$ ). Per-player score standard deviations are estimated from published distributions for elite amateur stroke play ( $\sigma_{\text{Oakmont}} \approx 3.8$ ,  $\sigma_{\text{Longue Vue}} \approx 3.1$ ; Broadie, 2012). Robustness to SD assumptions is reported.

#### 3.3.4 Cross-Gender Comparison and Within-Venue Variance

The consecutive staging of Women's and Men's Opens at Pinehurst No. 2 in 2014 constitutes a controlled natural experiment for cross-gender difficulty comparison; we report the observed 36-hole cut differential. For venues with two or more modern era visits, we compute within-venue FSA+/- variance ( $s^2$ ) as a measure of condition-sensitivity, recognizing that formal F-tests for equality of variance are underpowered with  $n = 2$  per venue.

### 3.4 Venue Tier Classification Criteria

We classify venues into four tiers based on the totality of evidence across all available datasets:

- Tier 1 (Unconditional): Above-average FSA+/- across all available datasets — Men's Open, Women's Open, and Amateur — regardless of setup, weather, or era.
- Tier 2 (Strong case): Multi-dataset confirmation but limited sample size in one or more datasets.
- Tier 3 (Condition-sensitive): FSA+/- varies materially across visits, suggesting setup or weather are primary drivers rather than intrinsic architecture.
- Tier 4 (Easy): Adjusted difficulty data unambiguously below championship standard.

## 4. RESULTS

### 4.1 Modern Era Rankings: Men's U.S. Open (2010–2026)

Table 2 presents the complete modern era rankings by FSA+/-, ordered from most to least difficult. Tier assignments reflect the corrected groupings used in all formal statistical tests.

**Table 2. U.S. Open Field Scoring Average vs. Par, 2010–2026 — Ranked by Difficulty**

| Venue               | Year | FSA +/-      | Winner         | Win +/- | Yds/Par  | Tier                 | Notes                   |
|---------------------|------|--------------|----------------|---------|----------|----------------------|-------------------------|
| Merion (East)       | 2013 | <b>4.55</b>  | Justin Rose    | +1      | 6,996/70 | T2 — Strong case     | Hardest modern Open     |
| Shinnecock Hills    | 2018 | <b>4.39†</b> | Brooks Koepka  | +1      | 7,445/70 | T3 — Cond.-sensitive | Setup error (\$3.1)     |
| Winged Foot (West)  | 2020 | <b>4.07</b>  | B. DeChambeau  | −6      | 7,477/70 | T1 — Unconditional   | DeChambeau outlier      |
| Pebble Beach        | 2010 | <b>3.98</b>  | G. McDowell    | −1      | 6,816/71 | T3 — Cond.-sensitive | Wind-driven             |
| Oakmont CC          | 2025 | <b>3.82</b>  | J.J. Spaun     | −1      | 7,255/70 | T1 — Unconditional   | Sole sub-par (72 holes) |
| Olympic Club (Lake) | 2012 | <b>3.47</b>  | Webb Simpson   | +1      | 7,170/70 | T2 — Strong case     | Classic Olympic grind   |
| Chambers Bay        | 2015 | <b>3.45</b>  | Jordan Spieth  | −5      | 7,618/70 | T3 — Cond.-sensitive | Links-style setup       |
| Oakmont CC          | 2016 | <b>3.33</b>  | Dustin Johnson | −4      | 7,255/70 | T1 — Unconditional   | See §4.3                |
| Pinehurst No. 2     | 2014 | <b>3.09</b>  | Martin Kaymer  | −9      | 7,562/70 | T3 — Cond.-sensitive | Same wk as Women's Open |
| Pinehurst No. 2     | 2024 | <b>2.78</b>  | B. DeChambeau  | −6      | 7,500/70 | T3 — Cond.-sensitive | Trending more benign    |

| Venue             | Year | FSA +/- | Winner         | Win +/- | Yds/ Par | Tier                 | Notes                |
|-------------------|------|---------|----------------|---------|----------|----------------------|----------------------|
| Shinnecock Hills  | 2026 | 2.68    | Wyndham Clark  | -4      | 7,445/70 | T3 — Cond.-sensitive | See §5.3             |
| The Country Club  | 2022 | 2.60    | M. Fitzpatrick | -6      | 7,264/70 | T3 — Cond.-sensitive | Classic Brookline    |
| Torrey Pines (S)  | 2021 | 2.27    | Jon Rahm       | -6      | 7,652/71 | T3 — Cond.-sensitive | Long, manageable     |
| LACC (North)      | 2023 | 1.76    | Wyndham Clark  | -10     | 7,573/70 | T4 — Easy            | 62s shot in Round 1  |
| Congressional (B) | 2011 | 1.64    | Rory McIlroy   | -16     | 7,574/71 | T4 — Easy            | McIlroy dominant     |
| Pebble Beach      | 2019 | 1.22    | Gary Woodland  | -13     | 6,816/71 | T4 — Easy            | Calm; benign setup   |
| Erin Hills        | 2017 | 1.14    | Brooks Koepka  | -16     | 7,741/72 | T4 — Easy            | Widest fairways ever |

† 2018 Shinnecock partially inflated by USGA setup error Saturday afternoon (see §3.1). FSA +/- = field scoring average vs. par. Tier 1+2 (n=5): Oakmont ×2, Winged Foot 2020, Merion 2013, Olympic 2012. Tier 3 (n=7): Shinnecock, Pebble 2010, Chambers Bay, Pinehurst ×2, Country Club, Torrey Pines. Tier 4 (n=4): LACC, Congressional, Pebble 2019, Erin Hills. Source: Data Golf ([datagolf.com](http://datagolf.com)); USGA records.

The primary finding of Table 2 is that Merion Golf Club (East Course) in 2013 produced the highest adjusted field average of the modern era at +4.55, exceeding Oakmont Country Club (2025: +3.82; 2016: +3.33) and Shinnecock Hills (2018: +4.39, partially upward-biased by setup error). This result challenges the widespread assumption that Oakmont is the premier U.S. Open difficulty test.

The three lowest-difficulty observations — Erin Hills 2017 (+1.14), Pebble Beach 2019 (+1.22), and LACC 2023 (+1.76) — were not national championships in any meaningful difficulty sense. The USGA's stated objective of identifying a winner near even par was not achieved in any of these years.

The Pebble Beach variance deserves specific attention. Pebble Beach is widely documented as possessing some of the smallest greens of any major championship venue — small, exposed targets above ocean bluffs creating extreme sensitivity to wind. The 2010 edition (+3.98) and 2019 edition (+1.22) represent a 2.76-stroke swing at the same course, the largest within-venue variance in the dataset. This architectural characteristic — small greens that are punishing in wind but accessible in calm — explains the variance without implying inconsistency in USGA setup.

## 4.2 Formal Statistical Results

### 4.2.1 Welch's Two-Sample t-Test: Tier 1+2 vs. Tier 3+4

Tier 1+2 venues (Oakmont CC ×2, Winged Foot West, Merion East, Olympic Club Lake;  $n = 5$  modern observations) produced mean  $FSA_{+/-} = 3.848$  ( $SD = 0.488$ ). Tier 3+4 venues (all remaining venues, now including Shinnecock Hills 2026;  $n = 12$ ) produced mean  $FSA_{+/-} = 2.583$  ( $SD = 1.039$ ). Welch's two-sample t-test confirmed a statistically significant difference between groups:  $t(14.5) = 3.408$ ,  $p = 0.0041$  (two-tailed). The effect size was very large (Cohen's  $d = 1.558$ ), indicating that the two venue groups differ by more than 1.5 standard deviations in adjusted field scoring average. Results are summarized in Table 3.

**Table 3. Welch's Two-Sample t-Test: Tier 1+2 vs. Tier 3+4 Venues**

| Group                       | n  | Mean<br>$FSA_{+/-}$ | SD    | t            | df   | p              | Cohen's d                 |
|-----------------------------|----|---------------------|-------|--------------|------|----------------|---------------------------|
| Tier 1+2 — high difficulty  | 5  | <b>3.848</b>        | 0.488 | <b>3.408</b> | 14.5 | <b>0.0041*</b> | <b>1.558 (very large)</b> |
| Tier 3+4 — lower difficulty | 12 | 2.583               | 1.039 | —            | —    | —              | —                         |

*Two-tailed Welch's t-test (unequal variances assumed). \*\*  $p < 0.01$ . Cohen's d: small  $\geq 0.2$ , medium  $\geq 0.5$ , large  $\geq 0.8$ , very large  $> 1.2$  (Cohen, 1988).  $FSA_{+/-}$  = field scoring average vs. par. Tier 1+2: Oakmont CC (2016, 2025), Winged Foot West (2020), Merion East (2013), Olympic Club Lake (2012). All computations verified in supplementary Python script (`scipy.stats.ttest_ind, equal_var=False`).*

### 4.2.2 One-Way ANOVA: $FSA_{+/-}$ Across All Four Venue Tiers

One-way ANOVA confirmed a highly significant difference in mean  $FSA_{+/-}$  across the four proposed venue tiers:  $F(3, 13) = 12.001$ ,  $p = 0.0005$ . Tier means: Tier 1 (Unconditional) = 3.740, Tier 2 (Strong case) = 4.010, Tier 3 (Condition-sensitive) = 3.155, Tier 4 (Easy) = 1.440. Estimated eta-squared  $\eta^2 = 0.735$ , indicating that approximately 73.5% of the total variance in  $FSA_{+/-}$  is explained by tier membership. This is a very large effect and confirms that the four-tier classification captures systematic rather than arbitrary variance in venue difficulty. Results are presented in Table 4.

**Table 4. One-Way ANOVA —  $FSA_{+/-}$  Across Four Venue Tiers**

| Tier                          | n | Mean<br>$FSA_{+/-}$ | SD    | F             | df    | p                | Venues                                                  |
|-------------------------------|---|---------------------|-------|---------------|-------|------------------|---------------------------------------------------------|
| <b>Tier 1 — Unconditional</b> | 3 | <b>3.740</b>        | 0.376 | <b>12.001</b> | 3, 13 | <b>0.0005***</b> | WF West 2020; Oakmont 2016, 2025                        |
| <b>Tier 2 — Strong case</b>   | 2 | <b>4.010</b>        | 0.764 |               |       |                  | Merion 2013; Olympic 2012                               |
| Tier 3 — Condition-sensitive  | 8 | <b>3.155</b>        | 0.732 |               |       |                  | Shinnecock ×2; Pebble 2010; Chambers Bay; Pinehurst ×2; |



| Tier          | n | Mean<br>FSA+/- | SD    | F | df | p | Venues                                       |
|---------------|---|----------------|-------|---|----|---|----------------------------------------------|
|               |   |                |       |   |    |   | Country Club; Torrey Pines                   |
| Tier 4 — Easy | 4 | <b>1.440</b>   | 0.306 |   |    |   | LACC; Congressional; Pebble 2019; Erin Hills |

\*\*\*  $p < 0.001$ . One-way ANOVA (*scipy.stats.f\_oneway*). Eta-squared  $\eta^2 = SS_{\text{between}} / SS_{\text{total}} = 0.735$  (73.5% of variance explained by tier).  $F$  and  $p$  reported for the omnibus test; blank cells in rows 2–4 indicate group summary statistics, not separate test statistics. Tier 3 includes Shinnecock Hills 2018 and 2026 (2018 setup-inflation caveat noted in §3.1).

#### 4.2.3 2021 U.S. Amateur Natural Experiment: Causal Identification

The quasi-experimental co-host design of the 2021 U.S. Amateur at Oakmont Country Club and Longue Vue Club provides the strongest causal evidence for architecture-driven difficulty in the dataset. Under random venue assignment, the mean Round 1 score at Oakmont ( $\bar{x} = 77.16$ ,  $\hat{\sigma} \approx 3.8$ ,  $n \approx 78$ ) was 5.47 strokes higher than the simultaneous mean at Longue Vue Club ( $\bar{x} = 71.69$ ,  $\hat{\sigma} \approx 3.1$ ,  $n \approx 78$ ). Welch's two-sample t-test yields  $t(148) \approx 9.851$ ,  $p < 0.0001$  (one-tailed), Cohen's  $d \approx 1.577$ . Results are shown in Table 5.

This result carries three notable properties. First, it is produced without Open-specification rough ( $\approx 1.5$  inches vs.  $\approx 3.5$ – $4.0$  inches for the Open), attributing the difficulty differential to course design. Second, random assignment eliminates selection bias. Third, the result is robust to SD assumptions: with per-player SDs of 5.0 at both venues (conservative overestimate),  $t \approx 6.832$ ,  $p < 0.0001$ , confirming the conclusion regardless of estimation uncertainty.

**Table 5. 2021 U.S. Amateur — Round 1 Venue Difficulty (Welch's t-Test, Quasi-Experimental Design)**

| Venue                              | n            | Mean R1<br>score | $\hat{\sigma}^\dagger$ | Mean<br>diff. | t               | p                         | Cohen's d                                      |
|------------------------------------|--------------|------------------|------------------------|---------------|-----------------|---------------------------|------------------------------------------------|
| Oakmont CC (par 70; no Open rough) | $\approx 78$ | <b>77.16</b>     | $\approx 3.8$          | <b>+5.47</b>  | $\approx 9.851$ | <b>&lt;0.0001<br/>***</b> | <b><math>\approx 1.577</math> (very large)</b> |
| Longue Vue Club (par 70; co-host)  | $\approx 78$ | 71.69            | $\approx 3.1$          | —             | —               | —                         | —                                              |

\*\*\*  $p < 0.001$  (one-tailed;  $H_1: \mu_{\text{Oakmont}} > \mu_{\text{Longue Vue}}$ ).  $^\dagger$  Per-player SD estimated from USGA Amateur stroke play dispersion parameters (Broadie, 2012); individual-level data not publicly available.  $df \approx 148$  (Welch–Satterthwaite approximation). Robustness check: SD = 5.0 at both venues gives  $t \approx 6.832$ ,  $p < 0.0001$ . Random venue assignment confirmed by USGA co-host format.

#### 4.2.4 Cross-Gender Comparison: Pinehurst No. 2, 2014

The consecutive staging of the Women's and Men's U.S. Opens at Pinehurst No. 2 in June 2014 provides the most controlled cross-gender comparison in the dataset. The Women's 36-hole cut fell at +9, four strokes higher than the Men's cut of +5, despite the Women's layout being 913 yards shorter

(6,649 yards vs. 7,562 yards). Under a length-equals-difficulty model, this direction is counterintuitive and inconsistent with the null hypothesis that shorter yardage produces an equal or easier cut score. This is consistent with our architectural hypothesis that Pinehurst No. 2's Donald Ross-designed crowned, fall-away green complexes impose disproportionate difficulty on shorter-hitting players whose approaches originate from inside 150 yards. Individual-level approach shot data would be required for a formal test, which we identify as a priority for future research.

4.2.5 Within-Venue Variance: Condition-Sensitivity Quantified

For venues with two or more modern era observations, Table 6 reports within-venue FSA+/- variance as a quantitative measure of condition-sensitivity. Pebble Beach Golf Links shows variance  $s^2 = 3.809$  (range = 2.76 strokes), the highest in the dataset, consistent with its Tier 3 classification. Oakmont CC shows  $s^2 = 0.120$  (range = 0.49 strokes), the lowest, consistent with Tier 1. Shinnecock Hills, with the addition of the 2026 championship (§5.3), now has two modern-era observations (2018: +4.39, partially setup-inflated; 2026: +2.68) yielding  $s^2 = 1.462$  (range = 1.71 strokes) — intermediate between Oakmont's reproducible difficulty and Pebble Beach's high variance, and providing the first quantitative variance estimate for Shinnecock in this dataset.

Table 6. Within-Venue FSA+/- Variance — Venues with Multiple Modern Era Observations

| Venue            | Visits     | FSA+/- values | Mean  | Variance ( $s^2$ ) | Range | Tier (condition-sensitivity)                           |
|------------------|------------|---------------|-------|--------------------|-------|--------------------------------------------------------|
| Oakmont CC       | 2016, 2025 | 3.33, 3.82    | 3.575 | 0.120              | 0.49  | Tier 1 — Unconditional (lowest variance in dataset)    |
| Pinehurst No. 2  | 2014, 2024 | 3.09, 2.78    | 2.935 | 0.048              | 0.31  | Tier 3 — mean below era avg despite low variance       |
| Shinnecock Hills | 2018, 2026 | 4.39, 2.68    | 3.535 | 1.462              | 1.71  | Tier 3 — intermediate variance; see §5.3               |
| Pebble Beach     | 2010, 2019 | 3.98, 1.22    | 2.600 | 3.809              | 2.76  | Tier 3 — highest variance; small greens wind-dependent |

*Variance computed with Bessel's correction (ddof = 1; scipy.stats). Formal Levene or Bartlett tests for equality of variance require  $n \geq 3$  per group; reported descriptively here. Pebble Beach variance of 3.809 remains the highest in the dataset; Shinnecock's variance of 1.462, newly calculable with the 2026 observation, is the quantitative basis for retaining its Tier 3 classification with a more moderate condition-sensitivity profile than Pebble Beach.*

4.3 Cross-Gender Validation: U.S. Women's Open

Table 7 presents Women's Open data at shared venues, providing an independent difficulty signal at each course.

Table 7. U.S. Women's Open Results at Shared Venues

| Venue               | Year | Winner         | Win +/- | W Cut+/-   | M Cut/FSA+/- | $\Delta$ (W minus M)                  |
|---------------------|------|----------------|---------|------------|--------------|---------------------------------------|
| Oakmont CC          | 1992 | Patty Sheehan  | -4      | <b>+9</b>  | 8.0 (avg)    | +1.0 (women harder)                   |
| Oakmont CC          | 2010 | Paula Creamer  | -3      | <b>+10</b> | 8.0 (avg)    | +2.0 (women harder)                   |
| Pinehurst No. 2     | 2014 | Michelle Wie   | -2      | <b>+9</b>  | +5 (same wk) | +4.0 (women harder; counterintuitive) |
| Olympic Club (Lake) | 2021 | Yuka Saso      | -4      | <b>+6</b>  | 8.0 (avg)    | -2.0 (men slightly harder)            |
| Pebble Beach        | 2023 | Allisen Corpuz | -9      | <b>+6</b>  | 2.6 (avg)    | +3.4 (women harder)                   |

*W Cut = Women's Open 36-hole cut vs. par. M Cut/FSA = Men's Open average cut or FSA+/- across all visits.  $\Delta$  = Women's minus Men's figure; positive = women found venue harder. Sources: USGA Championship Records; Wikipedia; Golf Compendium. Olympic Club 2021 Women's rough trimmed from 3.25 to 2.75 inches pre-tournament (Note 5).*

Oakmont Country Club is confirmed as unconditionally difficult across gender: Women's Open cuts of +9 and +10 on a layout approximately 800 yards shorter than the Men's Open corroborate the Men's Open and Amateur evidence that Oakmont's difficulty is architectural. The Pinehurst 2014 result (Women's cut +9 vs. Men's cut +5 the same week, same course, same USGA crew) is the most controlled cross-gender data point in championship golf history and directly contradicts the simple hypothesis that shorter yardage implies easier scoring.

#### 4.4 Setup-Free Validation: U.S. Amateur Stroke Play

Table 8 presents Amateur data at shared venues, providing a setup-free architectural difficulty baseline.

**Table 8. U.S. Men's Amateur Stroke Play Data at Shared Venues**

| Venue            | Year | SP cut/avg       | Key finding                                                                                        |
|------------------|------|------------------|----------------------------------------------------------------------------------------------------|
| Oakmont CC       | 2003 | <b>~+6 est.</b>  | Only 3 players broke par in SP; co-host Pittsburgh FC far easier                                   |
| Oakmont CC       | 2021 | <b>+3 (cut)</b>  | R1 avg at Oakmont 77.16 vs. Longue Vue 71.69 ( $\Delta = 5.47$ ; $t \approx 9.85$ , $p < 0.0001$ ) |
| Winged Foot (W)  | 2004 | <b>~+2</b>       | Medalist Ryan Moore 139 (+3); elevated vs. typical Amateur medalist                                |
| Merion (East)    | 2005 | <b>~-5 (med)</b> | Medalist James Lepp 135 (-5); course more forgiving w/o Open setup                                 |
| Pebble Beach     | 2018 | <b>+5 (cut)</b>  | High cut; consistent with Open-year variance analysis (§4.2.5)                                     |
| Pinehurst No. 2  | 2019 | <b>+5 (cut)</b>  | Highest cut in 2016–2021 period; confirms condition-sensitivity                                    |
| Olympic Club (L) | 2021 | <b>+4.89</b>     | Co-host format (all players played both); Lake harder by ~0.95 strokes                             |

| Venue | Year | SP cut/avg             | Key finding                                                  |
|-------|------|------------------------|--------------------------------------------------------------|
|       | 5    | (Lake) / +3.94 (Ocean) | across two rounds (R1: 75.03 vs. 74.34; R2: 74.75 vs. 73.53) |

SP = stroke play qualifying. Sources: USGA Media Center; Golf Digest; Golf Bible; USGA Championship Records. Figures denoted 'est.' are author approximations. Statistical results for 2021 Oakmont from formal test in §4.2.3 and Table 5.

## 4.5 Summary of All Formal Statistical Tests

**Table 9. Summary of Statistical Tests — All Formal Results**

| # | Test                        | Comparison                           | Statistic                      | p-value                  | Effect size           | Conclusion                                   |
|---|-----------------------------|--------------------------------------|--------------------------------|--------------------------|-----------------------|----------------------------------------------|
| 1 | Welch's t-test (two-tailed) | Tier 1+2 vs. Tier 3+4<br>FSA+/-      | $t = 3.408$                    | <b>0.0041*</b><br>*      | $d = 1.558$           | Significant; high-diff. venues confirmed     |
| 2 | One-way ANOVA               | FSA+/- across 4 tiers                | $F = 12.001$                   | <b>0.0005*</b><br>**     | $\eta^2 = 0.735$      | Highly significant; tier framework validated |
| 3 | Welch's t-test (one-tailed) | Oakmont vs. Longue Vue 2021 Amateur  | $t \approx 9.851$              | <b>&lt;0.0001</b><br>*** | $d \approx 1.577$     | Causal: Oakmont architecture confirmed       |
| 4 | Directional (sign test)     | Pinehurst 2014 gender cut gap        | $\Delta = +4$ strokes          | N/A†                     | N/A†                  | Counterintuitive direction; Ross hypothesis  |
| 5 | Variance (descriptive)      | Pebble Beach $s^2$ vs. Oakmont $s^2$ | $s^2 = 3.809$ vs. <b>0.120</b> | N/A‡                     | Range $\Delta = 2.27$ | Condition-sensitivity confirmed numerically  |

\*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ . All inferential tests at  $\alpha = 0.05$ .  $\eta^2 = SS_{\text{between}} / SS_{\text{total}}$ . Cohen's  $d$ : small  $\geq 0.2$ , medium  $\geq 0.5$ , large  $\geq 0.8$ , very large  $> 1.2$ . † Single natural experiment; formal  $p$ -value requires individual-level data not publicly available. ‡ Formal Levene test requires  $n \geq 3$ ; reported descriptively. All computations reproducible via `supplementary_statistical_analysis.py`.

## 5. VENUE TIER CLASSIFICATION

### 5.1 Tier 1: Unconditional Difficulty

**Oakmont Country Club (Oakmont, PA).** Oakmont satisfies all criteria for Tier 1 classification across all four datasets. Across ten Men's Opens, FSA+/- has ranged from +3.33 to +5.70. Two Women's Opens (1992, 2010) produced cuts at +9 and +10. Six Men's Amateurs, including the 2021 event with its quasi-experimental 5.47-stroke venue differential ( $t \approx 9.851$ ,  $p < 0.0001$ ,  $d \approx 1.577$ ), confirm the architectural basis. Within-venue variance  $s^2 = 0.120$  across two modern Open visits is the lowest in the dataset, confirming reproducibility across conditions.

Henry Fownes' 1903 design features approximately 200 bunkers, severely contoured greens with near-zero flat putting surfaces, and fairways that reward precise positional play. The USGA consistently requires minimal artificial intervention at Oakmont to produce championship-level difficulty.

**Winged Foot Golf Club, West Course (Mamaroneck, NY).** Three Men's Opens have produced FSA+/- of +4.07, +5.00, and +6.99. The 2004 U.S. Amateur medalist score is consistent with above-average difficulty without Open setup. Women's Open data is pending (scheduled 2036). Classified Tier 1 on the strength of three-Open Men's consistency. A.W. Tillinghast's design philosophy prioritizes approach angle over distance — an era-independent characteristic.

## 5.2 Tier 2: Strong Case

**Olympic Club, Lake Course (San Francisco, CA).** Olympic Club has hosted five Men's Opens (1955, 1966, 1987, 1998, 2012), one Women's Open (2021), and one U.S. Amateur (2025) — a championship history exceeded among Tier 1 and Tier 2 venues only by Oakmont. Across all five Men's Opens combined, only four players in history have finished a single championship below par; the 1955 (Fleck d. Hogan) and 1966 (Casper d. Palmer) Opens are particularly notable for frontrunner collapses on the back nine, a pattern repeated in 1987 (Simpson d. Watson) and 1998 (Janzen d. Stewart) — four playoff or final-round reversals across five visits, a level of competitive volatility consistent with, though not formal statistical proof of, a course that punishes any lapse in precision under pressure. Verified field-level difficulty data exists for three of these modern-era events. The 2012 Men's Open produced FSA+/- = +3.47. The 2021 Women's Open produced a 36-hole cut of +6, achieved after the USGA preemptively reduced rough from 3.25 to 2.75 inches following player feedback (Note 5), making this figure a conservative lower bound on the course's unmodified difficulty. The 2025 U.S. Amateur, held in a co-host stroke-play format with the adjacent Ocean Course, produced a Lake Course field average of approximately +4.89 across two rounds (75.03 and 74.75 on a par-70 layout) versus +3.94 on the Ocean Course — the Lake Course playing nearly one stroke harder than its co-host with no Open setup whatsoever. This Amateur result, while not a randomized natural experiment in the manner of the 2021 Oakmont co-host design (every player played both courses rather than being assigned to one), is nonetheless the highest verified field-level difficulty figure recorded at Olympic Club in the modern era, exceeding even the 2012 Open. Sam Whiting's 1927 redesign — narrow, tree-lined fairways and severely sloping greens that reject all but

the most precisely struck approach — is classified Tier 2 rather than Tier 1 on the basis of the single modern Open data point and the rough-modified Women's Open figure; a second modern Men's Open, already anticipated in the USGA's rotation, would be the most direct path to Tier 1 reclassification.

**Merion Golf Club, East Course (Ardmore, PA).** Merion's 2013 Open ( $FSA_{+/-} = +4.55$ ) is the highest adjusted field average in the 2010–2026 dataset. The 2005 Men's Amateur medalist score of 135 (–5) suggests Merion's difficulty is substantially amplified by specific USGA setup choices rather than being fully unconditional. This distinguishes Merion from Oakmont and places it in Tier 2. The 2026 U.S. Women's Amateur at Merion will provide cross-gender evidence.

### 5.3 Tier 3: Condition-Sensitive

**Shinnecock Hills Golf Club (Southampton, NY).** Four documented Men's Opens (1986, 1995, 2004, 2018) plus the 126th U.S. Open (2026) show a wide range, from an even-par winning score (1995) to +4.39 field average (2018, partially inflated by setup failure). The 2004 and 2018 Opens both involved acknowledged USGA setup failures — a firm, fast 7th green in 2004 that became unplayable Sunday, and Saturday afternoon hand-watering in 2018 — establishing Shinnecock's reputation for setup volatility despite William Flynn's genuinely demanding 1931 architecture.

The 2026 championship, won by Wyndham Clark (276, –4; rounds of 64-69-70-73) by one stroke, produced  $FSA_{+/-} = +2.68$  across 416 completed rounds, with a standard deviation of 2.77 — comparable to a typical PGA Tour event and notably lower than Shinnecock's historical reputation for extreme variance would predict, with a 36-hole cut of +5. This is Shinnecock's lowest modern-era  $FSA_{+/-}$ , and combined with 2018 gives a two-observation within-venue variance of  $s^2 = 1.462$  (Table 6) — intermediate between Oakmont's reproducible severity and Pebble Beach's high variance. The USGA's stated 2026 setup philosophy — fairways averaging 48 yards wide, the widest in at least 50 years, with rough deliberately pulled back — appears to have moderated the extremes that characterized 2004 and 2018, at the cost of producing one of Shinnecock's more benign U.S. Opens by adjusted field average. Reports from the third round indicated the USGA firmed and quickened the greens overnight after the first two rounds were considered too gentle, suggesting that even within 2026, daily setup decisions continued to materially affect scoring — a within-tournament illustration of exactly the weather/setup confound discussed in §6.3.1. Shinnecock remains classified Tier 3: the architecture is demonstrably capable of producing Tier 1-caliber difficulty (2018) but has not done so reproducibly across its modern-era visits. Women's Open data remains pending (scheduled 2036).

**Pebble Beach Golf Links (Pebble Beach, CA).** Within-venue variance  $s^2 = 3.809$  (range 2.76 strokes) is the highest in the dataset and the quantitative basis for Tier 3 classification. Pebble's documented small greens create extreme sensitivity to wind. The 2023 Women's Open (cut +6) demonstrates that Pebble can test elite professionals in appropriate conditions.

**Pinehurst No. 2 (Pinehurst, NC).** Three Men's Opens have ranged from +2.78 to +4.50 (range = 1.72 strokes). Within-venue variance  $s^2 = 0.048$  is the second-lowest in the dataset, suggesting the course is not highly condition-sensitive — but mean FSA+/- = 2.935 falls below the era average of 2.973, keeping it in Tier 3. The 2014 cross-gender natural experiment (Women's cut +9 vs. Men's cut +5) is consistent with Donald Ross's design philosophy prioritizing approach angle over distance.

#### 5.4 Tier 4: Easy

Three events — Erin Hills 2017 (+1.14), Pebble Beach 2019 (+1.22), and LACC 2023 (+1.76) — are classified Tier 4. The Welch's t-test in §4.2.1 confirms this group is statistically distinct from Tier 1+2 venues ( $p = 0.0061$ ,  $d = 1.508$ ). All three produced winning scores of -10 or lower. This classification describes these specific events rather than the venues permanently.

## 6. DISCUSSION

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### 6.1 The Inflation Adjustment Framework

The analogy between era-adjusted FSA+/- and inflation-adjusted economic statistics merits elaboration. In macroeconomics, nominal GDP figures are not compared across decades without price-level adjustment. Similarly, a 1955 U.S. Open FSA+/- of +8.72 cannot be compared to a 2025 figure of +3.82 without adjusting for field quality, agronomic technology, and equipment. The pre-modern figures in Table 10 are presented as nominal indicators — directionally valid but not directly comparable in absolute terms to modern Data Golf-verified figures. This framework is more conservative than most golf analytics discourse, which frequently cites raw historical numbers as though directly equivalent to modern data.

### 6.2 Architectural Characteristics of Tier 1 Venues

The four-dataset validation reveals a consistent set of architectural properties shared by Tier 1 venues. Green complex severity: Tier 1 greens present genuinely difficult recovery options from short-

side misses. Approach angle premiums: Tier 1 courses reward specific positional play from the tee that is independent of rough height and therefore independent of USGA setup decisions. Era independence: Oakmont's FSA+/- pattern across events spanning 1927 to 2025 is more consistent than any exogenous weather or setup explanation could produce. The quasi-experimental 2021 Amateur data ( $t \approx 9.851$ ,  $d \approx 1.577$ ) is the strongest available causal evidence for this era-independence.

### 6.3 Statistical Limitations and Future Research

Several limitations constrain this analysis, four of which warrant detailed treatment because they bear directly on the causal interpretation of the venue tier classifications.

#### 6.3.1 *Weather as an Unmeasured Confound*

FSA+/- at the single-event level reflects a joint realization of architectural difficulty and that week's weather; it does not separate the two. With only one to three modern-era visits per venue, within-venue variance estimates (Table 6) are imprecise and cannot fully distinguish “this course is architecturally consistent” from “this course has not yet been observed under adverse conditions.” Pebble Beach's classification as condition-sensitive (Tier 3) is as much a statement about the limits of a two-observation variance estimate as it is a statement about the course's small, exposed greens. A rigorous weather control would incorporate on-site wind speed and direction — available from historical reanalysis products or USGA/broadcast records — as a round-level covariate predicting FSA+/- . Absent that covariate, the 2021 U.S. Amateur natural experiment at Oakmont (§4.2.3) is the only observation in this dataset where weather is held constant by design rather than assumed constant by inference, which is why it carries disproportionate evidential weight in the Tier 1 classification for that venue specifically.

The 2026 Shinnecock Hills Open illustrates this confound concretely within a single championship. Round-level scoring averages were 73.277 (Round 1,  $n = 155$ ), 72.252 (Round 2,  $n = 155$ ), 73.625 (Round 3,  $n = 72$ ), and 71.389 (Round 4,  $n = 72$ ) — equivalently +3.28, +2.25, +3.63, and +1.39 versus par 70. Round 3, played after the USGA firmed and quickened the greens overnight and sustained wind arrived, was 2.24 strokes harder than the final round and 0.86 strokes harder than the average of the first two rounds, despite all four rounds being contested on an identical, unchanged golf course. This within-tournament swing is smaller in magnitude than the 2.76-stroke between-tournament swing observed at Pebble Beach across separate Opens (Table 6), but it demonstrates the same underlying mechanism operating on a faster timescale: a single overnight setup and weather



change measurably altered field difficulty by roughly a stroke or more, independent of any change to the course's architecture. The aggregate 2026 FSA+/- of +2.68 — a figure that places Shinnecock 2026 in the middle of the modern era distribution rather than near the top, where its 2018 visit (+4.39) sits — is therefore a composite of at least two materially different difficulty regimes within the same week, a granularity that the single-number FSA+/- metric used throughout this paper cannot, by construction, distinguish.

### ***6.3.2 Cross-Era Field Quality and Talent Comparability***

The FSA+/- framework assumes that the collection of elite talent constituting “the field” is approximately comparable across the years being compared within the modern era (2010–2026); this assumption was asserted by analogy to inflation adjustment in §2.3 and §6.1 in earlier drafts of this paper. It is now tested directly. Using Data Golf's pre-cut, Round 1 field-strength index for each of the seventeen modern-era U.S. Opens — a continuous measure of the field's aggregate skill level independent of that week's realized scoring — we regress FSA+/- on field strength across all seventeen observations.

The relationship is weak and not statistically significant: Pearson  $r = 0.203$ ,  $p = 0.434$ ; simple linear regression yields  $R^2 = 0.041$  (field strength explains approximately 4% of the variance in FSA+/-), with slope = 2.370 (SE = 2.949,  $p = 0.434$ ). This null result is the favorable outcome for the validity of this paper's central methodology: it indicates that field quality, as measured by this index, is not a significant driver of the FSA+/- figures used throughout this analysis, supporting — rather than merely asserting by analogy — the inflation-adjustment assumption for the 2010–2026 period specifically.

This test also directly addresses the alternative explanation raised in §6.3.4 regarding Merion 2013. After regressing out field strength, Merion's residual FSA+/- is +1.527 — the second-largest positive residual in the dataset — indicating that Merion's extreme difficulty is not attributable to an unusually weak 2013 field; if anything, the model under-predicts Merion's difficulty even after accounting for field strength. Shinnecock Hills shows the largest positive residual in either of its two visits (2018: +1.426), while Erin Hills (−2.008) and Pebble Beach 2019 (−1.767) show the largest negative residuals, confirming that these Tier 4 “easy” classifications are not artifacts of unusually strong fields elevating the difficulty baseline — if anything, these events under-performed difficulty expectations net of field quality as well as in raw terms.

Two caveats apply to this test. First, the field-strength index used reflects the pre-cut Round 1 field only; it does not capture mid-tournament attrition (withdrawals, missed cuts) that could independently affect the back-nine-round average via a changing player population, a separate mechanism from the pre-tournament talent-comparability question this test addresses. Second, this analysis was conducted only for the modern era (2010–2026), where Data Golf maintains a consistent field-strength methodology; the concern remains substantially more acute for the pre-modern historical figures in Table 10, where field composition — a 1955 U.S. Open field of 1,522 entries including club professionals and regional amateurs — differs from a modern major championship field in ways that cannot be adequately captured by any retrospective index. This remains the basis for treating those figures as directional rather than comparable in absolute terms.

**Table 12. Field-Strength-Adjusted FSA+/-: Regression Residuals (n = 17)**

| Venue                     | Field Strength | FSA+/-      | Predicted | Residual      | Note                                     |
|---------------------------|----------------|-------------|-----------|---------------|------------------------------------------|
| <b>Merion 2013</b>        | -0.046         | <b>4.55</b> | 3.023     | <b>+1.527</b> | <i>2nd-largest positive residual</i>     |
| <b>Shinnecock 2018</b>    | -0.071         | <b>4.39</b> | 2.964     | <b>+1.426</b> | <i>Largest residual at this venue</i>    |
| <b>Pebble Beach 2010</b>  | -0.140         | <b>3.98</b> | 2.800     | <b>+1.180</b> |                                          |
| <b>Oakmont 2025</b>       | -0.077         | <b>3.82</b> | 2.950     | <b>+0.870</b> |                                          |
| <b>Oakmont 2016</b>       | -0.192         | <b>3.33</b> | 2.678     | <b>+0.652</b> |                                          |
| <b>Chambers Bay 2015</b>  | -0.120         | <b>3.45</b> | 2.848     | <b>+0.602</b> |                                          |
| <b>Olympic Club 2012</b>  | -0.053         | <b>3.47</b> | 3.007     | <b>+0.463</b> |                                          |
| <b>Winged Foot 2020</b>   | +0.222         | <b>4.07</b> | 3.660     | <b>+0.410</b> | <i>Highest field strength in dataset</i> |
| <b>Pinehurst 2014</b>     | -0.080         | <b>3.09</b> | 2.943     | <b>+0.147</b> |                                          |
| <b>Pinehurst 2024</b>     | -0.090         | <b>2.78</b> | 2.919     | <b>-0.139</b> |                                          |
| <b>Shinnecock 2026</b>    | -0.057         | <b>2.68</b> | 2.998     | <b>-0.318</b> |                                          |
| <b>Country Club 2022</b>  | -0.086         | <b>2.60</b> | 2.930     | <b>-0.330</b> |                                          |
| <b>Torrey Pines 2021</b>  | -0.120         | <b>2.27</b> | 2.849     | <b>-0.579</b> |                                          |
| <b>Congressional 2011</b> | -0.199         | <b>1.64</b> | 2.660     | <b>-1.020</b> |                                          |
| <b>LACC 2023</b>          | -0.108         | <b>1.76</b> | 2.877     | <b>-1.117</b> |                                          |
| <b>Pebble Beach 2019</b>  | -0.061         | <b>1.22</b> | 2.987     | <b>-1.767</b> | <i>2nd-largest negative residual</i>     |
| <b>Erin Hills 2017</b>    | +0.007         | <b>1.14</b> | 3.148     | <b>-2.008</b> | <i>Largest negative residual</i>         |

*Field strength: Data Golf pre-cut Round 1 field-strength index (continuous measure of field skill, independent scale; positive = stronger field). Predicted FSA+/- from OLS regression:  $FSA+/- = 3.132 + 2.370 \times \text{field\_strength}$  ( $R^2 = 0.041$ ,  $p = 0.434$ ). Residual =  $FSA+/- - \text{Predicted}$ ; large positive residuals indicate venues that played harder than field strength alone would predict. Source: author's analysis of supplementary field-strength data (us\_open\_strengths.csv).*

### **6.3.3 Tee Marker and Effective Course Rating Variation**

USGA Course Rating and Slope Rating are calculated per tee marker and can differ substantially — often several strokes of Course Rating and several points of Slope — between championship tees and forward tees at the same venue. FSA+/- as computed in this analysis is referenced to the posted par and yardage for the championship as a whole and does not account for daily or hole-by-hole tee marker movement, which the USGA routinely uses to vary effective length and difficulty across rounds independent of any change to the underlying architecture. This limitation bears most directly on two findings in this paper. First, the Merion 2013 result (§5.2): without as-played daily Course Rating data, this analysis cannot rule out that some portion of Merion's difficulty premium reflects unusually punishing tee marker decisions on specific holes that week rather than a stable architectural property reproducible across visits. Second, and more fundamentally, the Pinehurst No. 2 cross-gender comparison (§4.2.4): the 913-yard difference between the Men's and Women's tee sets is itself a tee marker difference, and the Course Rating/Slope differential between those two tee sets would predict some portion of the observed four-stroke cut-score gap on its own, independent of any green-complex architectural effect. This analysis has not isolated the residual gap that exceeds what the rating differential alone would predict, and the architectural interpretation offered in §4.2.4 should be read as a hypothesis consistent with the data rather than a finding that has fully excluded this alternative explanation. As-played daily Course Rating and Slope figures for historical U.S. Open championships are not, to the author's knowledge, systematically archived in publicly available form; obtaining them would likely require a direct request to the USGA's Course Rating and Handicapping department.

### **6.3.4 Observed-versus-Expected Scoring as a Superior Future Metric**

A more rigorous difficulty measure than FSA+/- would compare each player's observed score to an independently generated expected score — derived from either (a) the USGA Course Rating system's own definition of expected scratch-golfer performance, (b) player-level pre-tournament Strokes Gained: Total baselines of the kind maintained by Data Golf and the PGA Tour's ShotLink system, or (c) pre-tournament market-implied scoring expectations from sportsbook data. Any of these

approaches would simultaneously address the field-quality concern in §6.3.2 and the weather/setup concern in §6.3.1, because the expected-score baseline is generated independently of that week's realized outcome; the gap between observed and expected becomes a difficulty signal that has already absorbed who was playing and how they had been playing recently, isolating the portion of scoring attributable to the course and that week's conditions. Such data — particularly player-level pre-tournament baselines — was not available to the author at the time of this analysis. It is plausible that an observed-versus-expected reanalysis would alter some of the present findings rather than merely confirming them; the Merion 2013 result in particular has not been tested against the possibility that the 2013 field was, independent of the venue, in unusually poor pre-tournament form. The 2021 U.S. Amateur natural experiment at Oakmont is the one result in this dataset expected to be robust to such a reanalysis, because its random co-host assignment already controls for field quality by design.

Future research directions include: extending the field-strength regression in §6.3.2 to incorporate mid-tournament attrition effects and pre-modern observations as comparable indices become available; acquisition of historical wind speed and direction data to model weather as an explicit covariate; a request to USGA Course Rating staff for as-played daily tee marker and rating data at single-visit venues, particularly Merion; development of an observed-versus-expected scoring metric using Data Golf or ShotLink player baselines as the gold-standard refinement of FSA+/-; extension of the cross-dataset framework to British Open and Masters venues; and bootstrap confidence intervals for the Pinehurst 2014 cross-gender comparison if individual-level Women's Open round data can be obtained from the USGA.

## 7. CONCLUSIONS

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This analysis produces four primary conclusions, each supported by formal statistical evidence.

**First**, adjusted field scoring average, cross-validated across four independent datasets, is substantially superior to conventional metrics as a cross-era measure of U.S. Open venue difficulty. Welch's two-sample t-test confirms that high-difficulty venues produce significantly higher FSA+/- than lower-difficulty venues ( $t(14.5) = 3.408$ ,  $p = 0.0041$ ,  $d = 1.558$ ), and one-way ANOVA confirms the four-tier classification captures systematic variance ( $F(3,13) = 12.001$ ,  $p = 0.0005$ ,  $\eta^2 = 0.735$ ).

**Second**, Oakmont Country Club is the most consistently and unambiguously difficult venue in U.S. Open history, as confirmed by all four independent datasets. The 2021 U.S. Amateur quasi-

experiment ( $t \approx 9.851$ ,  $p < 0.0001$ ,  $d \approx 1.577$ ) provides causal — not merely correlational — evidence that Oakmont's difficulty is architectural. Its within-venue FSA+/- variance of  $s^2 = 0.120$  confirms this difficulty is reproducible regardless of conditions.

**Third**, the conventional assertion that Oakmont, Shinnecock Hills, and Winged Foot are the three hardest U.S. Open venues is partially inconsistent with the adjusted data. In the modern era, Merion Golf Club 2013 (+4.55) exceeds all three, and Shinnecock's evidence is compromised by documented setup variability. The statistically validated ranking places Oakmont and Winged Foot in confirmed Tier 1, Merion and Olympic Club in strong-evidence Tier 2, and Shinnecock in a genuinely difficult but condition-sensitive Tier 3.

**Fourth**, the cross-gender analysis at Pinehurst No. 2 in 2014 — Women's cut +9 vs. Men's cut +5 the same week on a course 913 yards shorter — provides a counterintuitive result consistent with the hypothesis that Donald Ross's design philosophy prioritizes approach angle over distance, imposing disproportionate difficulty on shorter-hitting players. This finding warrants further investigation through hole-by-hole scoring data.

APPENDIX: ALL-TIME HISTORICAL FIELD AVERAGES (WHERE DOCUMENTED)

Pre-modern figures are directional indicators only and excluded from formal tests due to field quality non-stationarity. See §2.3 and §3.1. This table, and the modern era dataset in Table 2, do not include every venue to have hosted a U.S. Open; venue inclusion criteria are addressed in Note 7.

Table 10. U.S. Open All-Time Field Scoring Averages (Men's, Where Documented)

| Year | Venue               | FSA+/- | Winner        | Win+/- | Era note                                             |
|------|---------------------|--------|---------------|--------|------------------------------------------------------|
| 1955 | Olympic Club (Lake) | 8.72   | Jack Fleck    | +7     | Pre-modern — directional only                        |
| 1951 | Oakland Hills (S)   | 7.23   | Ben Hogan     | +7     | Pre-modern — directional only                        |
| 1974 | Winged Foot (West)  | 6.99   | Hale Irwin    | +7     | Pre-modern — directional only                        |
| 2002 | Bethpage Black      | 4.90   | Tiger Woods   | -3     | Pre-modern — directional only; sole sub-par finisher |
| 2007 | Oakmont CC          | 5.70   | Angel Cabrera | +5     | Modern (Data Golf verified)                          |

| Year | Venue                 | FSA+/-       | Winner        | Win+/- | Era note                                           |
|------|-----------------------|--------------|---------------|--------|----------------------------------------------------|
| 1976 | Atlanta Athletic Club | <b>5.47</b>  | Jerry Pate    | -3     | <i>Pre-modern — directional only</i>               |
| 2006 | Winged Foot (West)    | <b>5.00</b>  | Geoff Ogilvy  | +5     | <i>Modern (Data Golf verified)</i>                 |
| 1978 | Cherry Hills CC       | <b>4.92</b>  | Andy North    | +1     | <i>Pre-modern — directional only</i>               |
| 1998 | Olympic Club (Lake)   | <b>4.60</b>  | Lee Janzen    | E      | <i>Pre-modern — directional only</i>               |
| 1999 | Pinehurst No. 2       | <b>4.50</b>  | Payne Stewart | -1     | <i>Pre-modern — directional only</i>               |
| 2013 | Merion (East)         | <b>4.55</b>  | Justin Rose   | +1     | <i>Modern (Data Golf) — hardest modern Open</i>    |
| 2018 | Shinnecock Hills      | <b>4.39†</b> | Brooks Koepka | +1     | <i>Modern — partial setup inflation (§3.1)</i>     |
| 2020 | Winged Foot (West)    | <b>4.07</b>  | B. DeChambeau | -6     | <i>Modern (Data Golf verified)</i>                 |
| 2025 | Oakmont CC            | <b>3.82</b>  | J.J. Spaun    | -1     | <i>Modern (Data Golf verified)</i>                 |
| 2009 | Bethpage Black        | <b>2.93</b>  | Lucas Glover  | -4     | <i>Pre-modern — heavy rain softened conditions</i> |
| 2026 | Shinnecock Hills      | <b>2.68</b>  | Wyndham Clark | -4     | <i>Modern (Data Golf verified) — see §5.3</i>      |

† Partially inflated by USGA setup error (§3.1). Pre-modern figures from GOLF Magazine, ESPN Stats & Information, USGA records. Modern era (2006+) from Data Golf (datagolf.com). Bethpage Black figures (2002, 2009) per Golf Digest analysis of PGA Tour-leading scoring averages those seasons. Pre-modern observations excluded from all formal inferential tests.

**Bethpage Black** (Farmingdale, NY), included in Table 10 for completeness, is notable as the only public golf course among the venues discussed in this paper. Its 2002 scoring average of +4.90 — with Tiger Woods the sole sub-par finisher — was the highest on the PGA Tour that season; its 2009 average of +2.93, also tour-leading that year, was substantially softened by continuous rain and multiple play suspensions. Both figures predate this paper's 2010 modern-era cutoff and are not included in Table 2 or any formal statistical test; the 2002 figure in particular, if era-adjustable, would rank among the highest in this dataset. Bethpage Black is a candidate for future-research tier classification on the same basis as Oakland Hills, Oak Hill, Inverness, Cherry Hills, and Hazeltine, discussed in §6.3.4.

## APPENDIX B: COURSE RATING AND SLOPE RATING REFERENCE TABLE

Table 11a presents U.S. Open venues' Course Rating and Slope Rating from the championship tees, sourced from the USGA National Course Rating Database (ncrdb.usga.org). This table is

explicitly descriptive: Course Rating and Slope are not substituted for FSA+/- in any formal statistical test in Sections 3–4. Its purpose is to let the reader compare each venue's static, normal-conditions difficulty rating against its empirically observed championship difficulty (Table 2), with the divergence between the two presented as an analytical finding in Table 11b.

**Table 11a. U.S. Open Venues — USGA Course Rating and Slope Rating (Championship Tees)**

| Venue                       | Tier   | Cours<br>e Rtg | Bogey<br>Rtg | Slop<br>e  | F9 Rtg | F9<br>Slop<br>e | B9 Rtg | B9<br>Slop<br>e | U.S. Open<br>Year(s) |
|-----------------------------|--------|----------------|--------------|------------|--------|-----------------|--------|-----------------|----------------------|
| <b>Torrey Pines (South)</b> | Tier 3 | <b>78.2</b>    | 105.3        | <b>148</b> | 38.9   | 145             | 39.3   | 150             | 2021                 |
| <b>Erin Hills</b>           | Tier 4 | <b>77.9</b>    | 104.5        | <b>145</b> | 38.8   | 143             | 39.1   | 147             | 2017                 |
| <b>Oakmont CC</b>           | Tier 1 | <b>77.7</b>    | 104.2        | <b>142</b> | 39.1   | 144             | 38.6   | 140             | 2016, 2025           |
| <b>Chambers Bay</b>         | Tier 3 | <b>77.6</b>    | 104.5        | <b>145</b> | 39.0   | 139             | 38.6   | 151             | 2015                 |
| <b>Congressional (Blue)</b> | Tier 4 | <b>77.4</b>    | 104.1        | <b>143</b> | 38.6   | 141             | 38.8   | 145             | 2011                 |
| <b>Pinehurst No. 2</b>      | Tier 3 | <b>76.5</b>    | 101.9        | <b>138</b> | 38.0   | 136             | 38.5   | 140             | 2014, 2024           |
| <b>Winged Foot (West)</b>   | Tier 1 | <b>76.4</b>    | 102.7        | <b>140</b> | 37.9   | 138             | 38.5   | 142             | 2020                 |
| <b>The Country Club</b>     | Tier 3 | <b>76.3</b>    | 101.5        | <b>139</b> | 37.9   | 137             | 38.4   | 141             | 2022                 |
| <b>Pebble Beach</b>         | Tier 3 | <b>76.0</b>    | 102.7        | <b>145</b> | 38.3   | 146             | 37.7   | 144             | 2010, 2019           |
| <b>Olympic Club (Lake)</b>  | Tier 2 | <b>75.7</b>    | 102.2        | <b>143</b> | 38.1   | 143             | 37.6   | 142             | 2012                 |
| <b>LACC (North)</b>         | Tier 4 | <b>75.6</b>    | 101.4        | <b>141</b> | 37.5   | 139             | 38.1   | 143             | 2023                 |
| <b>Shinnecock Hills</b>     | Tier 3 | <b>74.7</b>    | 101.7        | <b>145</b> | 37.7   | 139             | 37.0   | 152             | 2018                 |
| <b>Merion (East)</b>        | Tier 2 | <b>74.0</b>    | 100.8        | <b>145</b> | 36.3   | 142             | 37.7   | 147             | 2013                 |

*Sorted by Course Rating, descending. Source: USGA National Course Rating Database ([ncrdb.usga.org](http://ncrdb.usga.org)), championship/longest tee configuration, verified by author. Rtg = Rating. F9/B9 = front nine / back nine. Bogey Rating reflects expected scoring for a bogey-level player; Course Rating reflects expected scoring for a scratch player, both under normal (non-championship) conditions.*

Table 11b ranks the ten modern-era venues with both a Course Rating and an FSA+/- figure (Table 2) by each measure independently, and reports the gap between the two rankings. A large positive divergence means a venue's championship performance was considerably harder than its static rating would predict; a large negative divergence means the opposite.

**Table 11b. Course Rating Rank vs. FSA+/- Rank (n = 10 venues with both measures)**

| Venue                       | Course Rtg | CR Rank | FSA+/-      | FSA Rank | Divergence |
|-----------------------------|------------|---------|-------------|----------|------------|
| <b>Merion (East)</b>        | 74.0       | 10      | <b>4.55</b> | 1        | <b>+9</b>  |
| <b>Shinnecock Hills</b>     | 74.7       | 9       | <b>4.39</b> | 2        | <b>+7</b>  |
| <b>Olympic Club (Lake)</b>  | 75.7       | 7       | <b>3.47</b> | 4        | <b>+3</b>  |
| <b>Winged Foot (West)</b>   | 76.4       | 5       | <b>4.07</b> | 3        | <b>+2</b>  |
| <b>The Country Club</b>     | 76.3       | 6       | <b>2.60</b> | 6        | <b>+0</b>  |
| <b>LACC (North)</b>         | 75.6       | 8       | <b>1.76</b> | 8        | <b>+0</b>  |
| <b>Chambers Bay</b>         | 77.6       | 3       | <b>3.45</b> | 5        | <b>-2</b>  |
| <b>Torrey Pines (South)</b> | 78.2       | 1       | <b>2.27</b> | 7        | <b>-6</b>  |
| <b>Congressional (Blue)</b> | 77.4       | 4       | <b>1.64</b> | 9        | <b>-5</b>  |
| <b>Erin Hills</b>           | 77.9       | 2       | <b>1.14</b> | 10       | <b>-8</b>  |

*Divergence = CR Rank – FSA Rank, computed within the full  $n = 10$  set of modern-era venues having both measures. Positive (red) = played harder in championship competition than static rating predicts; negative (blue) = played easier; zero (gray) = ranks align. Source: Table 2 (FSA+/-) and Table 11a (Course Rating).*

The divergence pattern in Table 11b directly illustrates this paper's central methodological argument. **Merion (East)** has the lowest Course Rating of any venue in this comparison (74.0, ranked 10th of 10) yet produced the highest FSA+/- in the modern era (+4.55, ranked 1st) — a divergence of +9 rank positions, the largest in the dataset. Under its static rating alone, Merion would be predicted to be the easiest test in this group; in practice, it was the hardest. This is consistent with the architectural argument developed in §5.2: Merion's difficulty derives from green complex severity and approach angle demands that the Course Rating system, calibrated to normal playing conditions, does not capture at the levels generated by championship setup.

**Shinnecock Hills** shows the second-largest positive divergence (+7). **Erin Hills**, **Torrey Pines (South)**, and **Congressional (Blue)** show the largest negative divergences (-8, -6, and -5 respectively): all three carry Course Ratings among the highest in the dataset yet produced three of the four lowest FSA+/- figures in the modern era. This pattern suggests that raw length and a high static rating do not reliably predict championship-week difficulty once USGA setup, weather, and field approach strategy are introduced, lending additional empirical support to this paper's claim in §5 and §6.1 that FSA+/- — not Course Rating — is the more reliable measure of realized championship difficulty. This divergence analysis should not be read as a criticism of the USGA Course Rating System, which is explicitly designed to measure expected scoring under normal play and was never intended to predict championship-week outcomes under modified setup conditions.



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## Notes

- 1 Field scoring average versus par (FSA+/-) is defined as the mean score across all players who completed all four rounds, relative to course par. Players who withdrew or were disqualified are excluded.
  - 2 The term 'adjusted' in this paper refers to within-era normalization implicit in anchoring FSA+/- to each era's competitive standard, not to the PGA Tour's ShotLink-based strokes-gained adjusted scoring average.
  - 3 The 2018 Saturday afternoon round at Shinnecock Hills was halted during play on multiple occasions. The USGA setup crew watered the 12th, 15th, and 17th greens between groups. Saturday afternoon scores are included in the official FSA+/- figure reported here.
  - 4 Ludvig Åberg won the 2024 Farmers Insurance Open at Torrey Pines in his PGA Tour debut season and has subsequently achieved multiple additional victories.
  - 5 The Olympic Club 2021 Women's Open rough was trimmed from the originally planned 3.25 inches to 2.75 inches prior to competition following Tuesday practice round feedback. This setup adjustment makes direct comparison to Men's Open figures approximate rather than exact.
  - 6 Per-player score standard deviations for the 2021 U.S. Amateur analysis ( $\sigma_{\text{Oakmont}} \approx 3.8$ ,  $\sigma_{\text{Longue Vue}} \approx 3.1$ ) are estimated from published distributions for USGA championship stroke play events. The conclusion ( $p < 0.0001$ ) is robust to reasonable variation: with SDs of 5.0 at both venues,  $t \approx 6.832$  and  $p < 0.0001$ .
  - 7 Venue inclusion in this paper's primary dataset (Table 2) is governed by a single criterion: whether the venue hosted a Men's U.S. Open within the 2010–2026 modern era window for which Data Golf maintains verified field scoring average data. This is a data-availability criterion, not a judgment about a venue's architectural merit or historical importance. Several venues with substantial U.S. Open history are consequently absent from Table 2 and discussed only in the historical appendix (Table 10) or not at all, including Bethpage Black (2002, 2009 — both pre-dating the study window), Southern Hills, Medinah, Baltusrol, Oakland Hills, Oak Hill, Inverness, Cherry Hills, and Hazeltine National. Bethpage Black, Oakland Hills, Oak Hill, Inverness, Cherry Hills, and Hazeltine are identified in §6.3.4 as priority candidates for tier classification in future work extending this paper's window to include additional modern-era championships at those sites (e.g., Oakland Hills 2034, Hazeltine's prior Opens) or, where only pre-modern data exists, applying a field-strength-adjusted reanalysis per §6.3.2. Their absence from the present tier classification (§5) reflects a data limitation rather than a substantive finding about those venues' difficulty.
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## References

- Broadie, M. (2012). Assessing golfer performance on the PGA TOUR. *Interfaces*, 42(2), 146–165.
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.
- Cornish, G. S., & Whitten, R. E. (1993). *The Architects of Golf*. HarperCollins.
- Data Golf. (2025). U.S. Open Championship results database. Retrieved July 2026, from <https://datagolf.com/past-results/majors/26>
- Data Golf. (2026). U.S. Open field strength index by tournament round, 2010–2026 [Data set]. Author's compilation.
- Doak, T. (1992). *The Anatomy of a Golf Course*. Lyons & Burford.
- Hurdzan, M. J. (2004). *Golf Course Architecture: Evolutions in Design, Construction, and Restoration Technology*. Sleeping Bear Press.
- National Golf Foundation. (2024). *Participation and economic impact of golf in the United States*. NGF Research.
- Price, R. (1989). *The World of Golf*. Random House.
- Ruxton, G. D. (2006). The unequal variance t-test is an underused alternative to Student's t-test and the Mann–Whitney U test. *Behavioral Ecology*, 17(4), 688–690.
- United States Golf Association. (2026). 126th U.S. Open Championship: Shinnecock Hills Golf Club — media guide and setup notes.

- United States Golf Association. (2025). 125th U.S. Open Championship: Oakmont Country Club — official results and media guide.
- United States Golf Association. (2021). 121st U.S. Amateur Championship: Oakmont Country Club and Longue Vue Club — official results.
- United States Golf Association. (2019). USGA Course Rating and Slope System (4th ed.).
- United States Golf Association. (2026). National Course Rating Database. [ncrdb.usga.org](https://ncrdb.usga.org).
- Official World Golf Ranking. (2026). OWGR methodology and historical field strength data. [owgr.com](https://owgr.com).
- Ward, J., & Godwin, J. (2014). *The Golf Courses of the World*. Mitchell Beazley.
- Wikipedia contributors. (2025). U.S. Women's Open Championship results by venue. Wikipedia, The Free Encyclopedia.

## Data Availability

Modern era Men's Open field scoring averages (2010–2026) are publicly available at <https://datagolf.com/past-results/majors/26>. Historical field averages are reproduced from published sources cited above. U.S. Amateur stroke play data is available via USGA Media Center press releases archived at [usga.org](https://usga.org). Women's Open cut scores are available via the USGA Championship Records database and Wikipedia. No proprietary or restricted datasets were used. The supplementary Python script (`statistical_analysis.py`) and four CSV data files reproduce all formal statistical results reported in Tables 3–6 and Table 9. Data compilation files are available from the author upon request.

## Competing Interests

The author declares no competing financial interests. The author is a member of the Olympic Club (San Francisco, California), a venue discussed and classified in this analysis. This affiliation is disclosed in the interest of full transparency; no financial relationship exists between the author and any venue, organization, or data provider discussed in the manuscript. Data Golf, whose verified field scoring averages are cited as a primary data source, was not consulted in the preparation of this research. This research received no external funding.

## Artificial Intelligence Disclosure

In accordance with De Gruyter's guidelines for authors, the author discloses that AI-assisted writing tools (Claude, developed by Anthropic) were used in the preparation of this manuscript, including assistance with drafting, structural organization, data table compilation, and document formatting. All intellectual contributions, analytical decisions, statistical computations, and claims of originality are the author's own. The AI tool did not generate the underlying data, perform the statistical computations

reported in Tables 3–6 and Table 9, or make any substantive analytical judgments. Statistical results were independently computed and verified by the author using Python (scipy.stats) prior to inclusion in the manuscript.

**Funding**

This research received no external funding from any government agency, funding body, commercial entity, or private institution.